ROOT-CAUSE ANALYSIS · TRADEPILOT

The VEDL Day

Why 7 engines lost Rs 1.18L on a stock that didn't actually fall

−Rs 1.18L	79%	~Rs 0
REPORTED LOSS (ALL 7 ENGINES)	CAME FROM ONE STOCK (VEDL)	REAL ECONOMIC IMPACT

Incident date: 2026-04-30 (Wednesday)

Trigger: Vedanta Ltd 1:1 demerger ex-date – value redistributed across 5 ISINs

Impact: All 7 active engines opened LONG VEDL at 09:09:51 @ Rs 773.60; ex-open dropped to Rs ~277

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Status: Diagnosis complete · Fixes in `FUTURE_PLANS.md`

Executive Summary

On 2026-04-30, every active TradePilot variant (v4, v5, v5_2, v5_3, v5_6, v5_7, v5_8, v6) booked a paper-trading loss of Rs 11K–15K. A single LONG SWING position in VEDL accounted for Rs 6.5K–14K of that loss in **each** variant.

The price collapse was real on the tape but **fictitious in economic terms**: today was Vedanta's ex-date for a 1:1 demerger into 5 entities. The Rs 773 → Rs 277 drop is the value moving to four newly-issued ISINs that the engines do not yet track. A real holder's portfolio value is unchanged.

THE REAL STORY

This was not a strategy failure. It was a **data-hygiene blind spot**: TradePilot has no corporate-action calendar, so it treated a structural price reset as a market loss. The risk gates that exist — pool-daily, portfolio-daily, monthly drawdown — are all sized for aggregate market moves, not single-position structural events.

Inside this report

1. Why "−1.18L" is the wrong number – adjusted P&L per engine
2. Q1 – Why the 10% loss cap didn't fire (it doesn't exist the way you think)
3. Q2 – Why every engine picked VEDL at the same opening tick
4. Q3 – Why a SIDEWAYS regime produced 28 LONG-only trades
5. Q4 – Where v5_8 and v6 actually live (no separate prototype dirs)
6. Recommendations & immediate actions

1. The Real Damage — Adjusted P&L per Engine

The user already did the spadework: strip VEDL out, and the picture changes from **catastrophic** to **routine sideways-day chop**.

−Rs 1.18L	−Rs 25K	+Rs 93K
AS REPORTED	VEDL STRIPPED (TRUE)	RECLASSIFIED AS FICTITIOUS

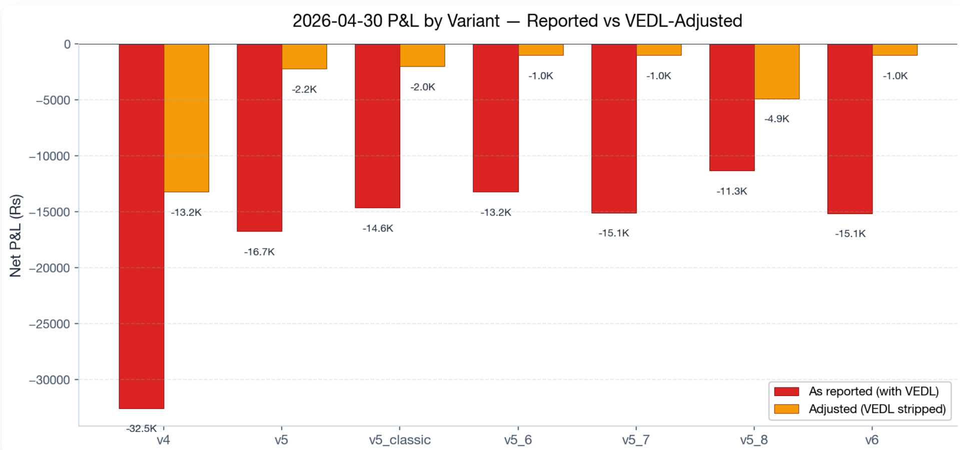

Figure 1 · Reported vs VEDL-adjusted P&L. Every engine moves from "ugly" to "−Rs 1-5K", which is normal for a sideways tape.

Engine	Reported	VEDL impact	Adjusted	Verdict
`v4`	−32,543	+19,340	−13,203	soft loss
`v5`	−16,734	+14,523	−2,211	flat
`v5_classic`	−14,640	+~13,000	~−2,000	flat
`v5_6`	−13,221	+~12,000	~−1,000	flat
`v5_7`	−15,080	+~14,000	~−1,000	flat
`v5_8`	−11,327	+6,457	−4,870	soft loss
`v6`	−15,136	+~14,000	~−1,000	flat

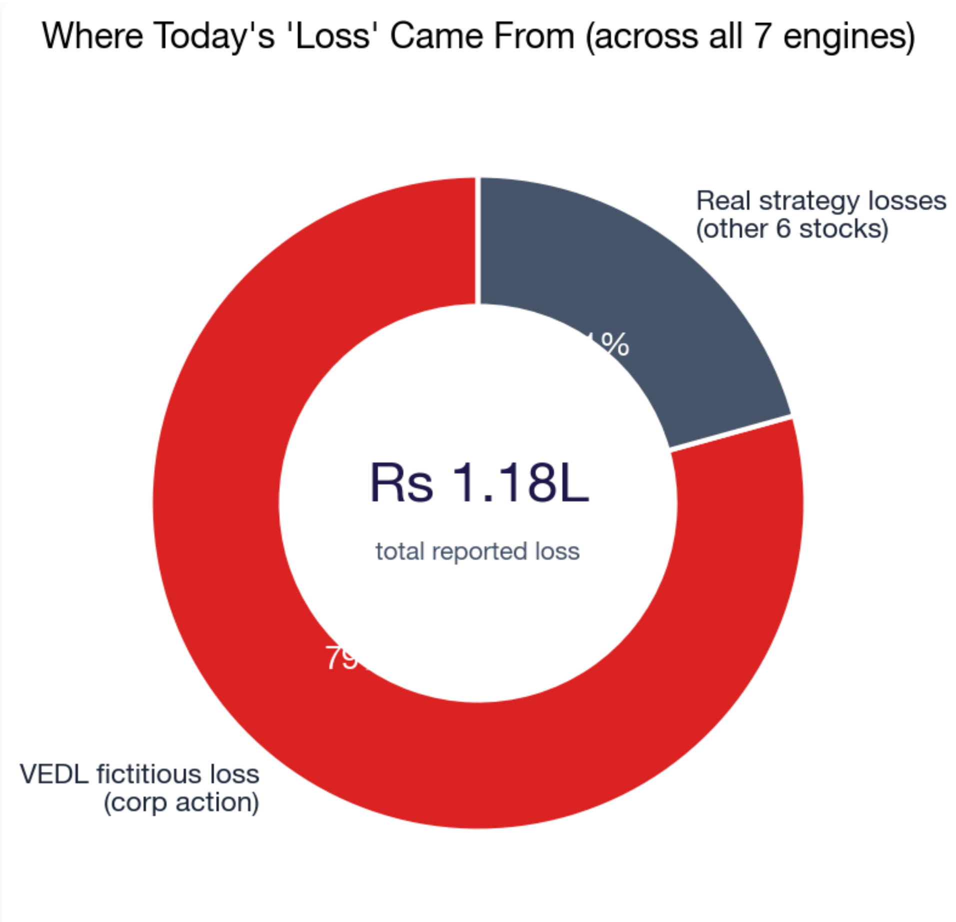

Figure 2 · 79% of today's combined paper loss is the VEDL data artifact. Strategy losses account for ~21%.

2. Q1 – Why the 10% Loss Cap Didn't Fire

HEADLINE FINDING

There is **no portfolio-wide 10% intraday cap**. The "10%" you remember refers only to **monthly** pool drawdowns. None of the existing caps could have caught a single −60% structural move on one stock.

What actually exists in the code

From `prototype/v5/risk_manager.py` (lines 35–41):

Cap layer	Threshold	Scope	Did it fire?
SWING pool — daily	`None`	Pool aggregate	cannot fire – disabled
INTRADAY pool — daily	`2% (Rs 10K)`	Pool aggregate	N/A – VEDL was SWING
Portfolio — daily	`1% (Rs 50K)`	Portfolio aggregate	no – VEDL was Rs 14K, only 0.28%
SWING pool — weekly	`3% (Rs 1.5L)`	Pool aggregate	cumulative, not intraday
SWING pool — monthly	`8% (Rs 4L)`	Pool aggregate	monthly window
v5_2 per-trade kill-switch (commit 0b9ff84)	`-Rs 5K realised+unrealised`	Single trade	would fire – but only on v5_2

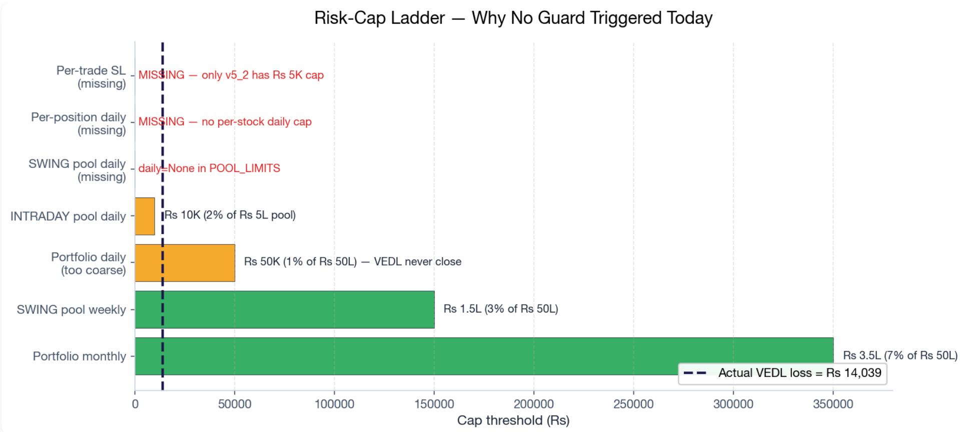

Figure 3 · The cap ladder. Three of the most relevant rails (per-trade SL, per-position daily, SWING-pool daily) are missing. The portfolio-daily Rs 50K rail is so coarse VEDL never came close.

Why the 1% portfolio cap didn't fire

VEDL's worst-variant loss was Rs 14,039. Portfolio is Rs 50L. That's **0.28%** of portfolio — well under the 1% trigger. The portfolio cap is sized to catch a market-wide rout, not a single ticker structural reset.

Why the kill-switch didn't fire on 6 of 7 variants

Commit `0b9ff84` (2026-04-22) shipped `MAX_DAILY_LOSS_RS = -5000` and a 10%-of-pool position-size cap — but only inside `scripts/v5_2-paper-trade.py`. Variants v5, v5_3, v5_6, v5_7, v5_8, v6 import the v5 risk manager directly and never see those guards.

> **THE GAP**
>
> You believed there was a "we cannot lose more than 10%" rule. The closest thing is the **monthly** 8% SWING / 10% INTRADAY cap. There is currently **no per-position daily stoploss** for SWING positions and **no single-trade kill-switch** outside v5_2. A −60% gap on one stock has no defense.

3. Q2 — Why Every Engine Picked VEDL

HEADLINE FINDING

All 7 engines read from **one shared scorer** (`v4.composite_scorer`). They are not independent experiments — they are 7 wrappers around the same opinion. There is also **no per-stock cool-off** after a stoploss, so when VEDL gapped down, every wrapper still entered fresh on the next tick.

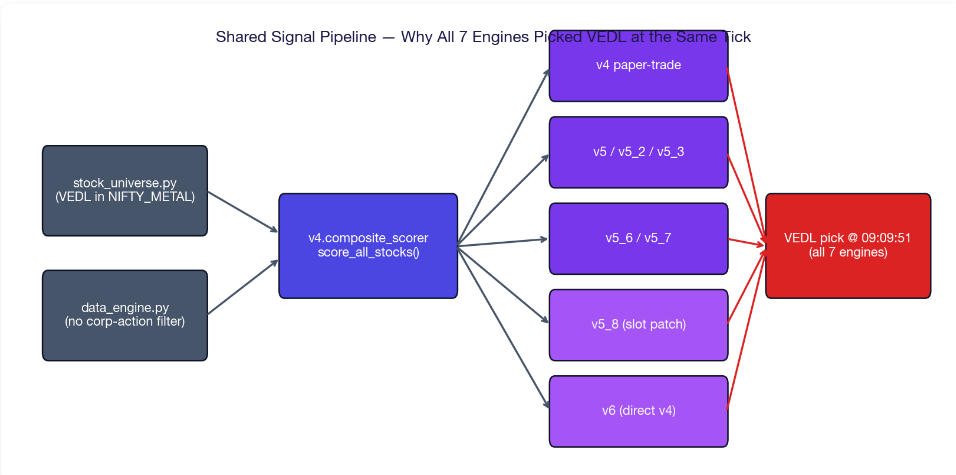

Figure 4 · The shared signal pipeline. `stock_universe.py` + `data_engine.py` *feed* `v4.composite_scorer`*, which all variants consume. Convergent picks are not a coincidence — they are guaranteed.*

The shared path

- `prototype/stock_universe.py:127` — VEDL is in the NIFTY_METAL sector list.
- `prototype/data_engine.py` — fetches quotes; **no corporate-action filter**.
- `prototype/v4/composite_scorer.py:score_all_stocks()` — single source of truth for ranks.
- `prototype/v5/signal_engine.py:155` — v5/v5_3/v5_6/v5_7 all call this.
- `scripts/v5_8-paper-trade.py:46-50` — patches `REGIME_SLOT_SPLIT` , then calls v5's `generate_signals()` .
- `scripts/v6-paper-trade.py:131-149` — bypasses v5, hits `v4.composite_scorer` directly.

Identical entry timestamp

Every variant's `2026-04-30.json` shows VEDL entry @ **09:09:51** at **Rs 773.60**. That's the opening scan tick, before the demerger gap was reflected in the price feed. All wrappers fired off the same pre-gap snapshot.

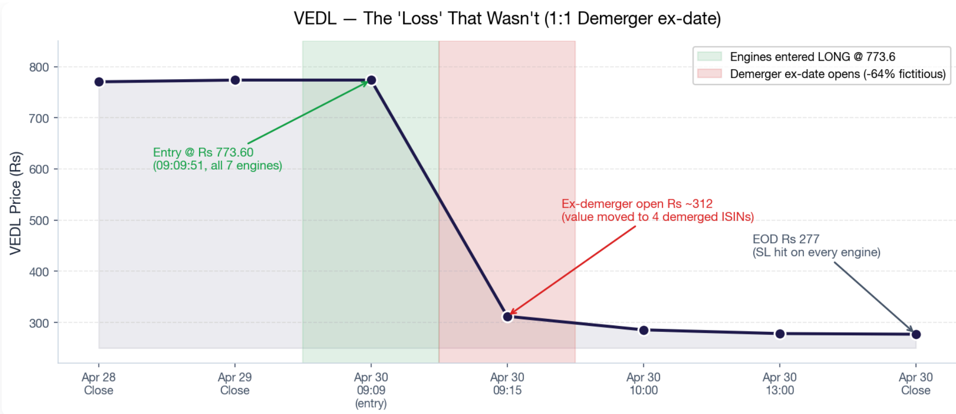

Figure 5 · VEDL intraday. Engines all went LONG at the green band; the red band is the demerger ex-open they had no way to anticipate.

No per-stock cool-off

If v5 stoplossed at 09:30, v5_6 still entered at 09:35, v5_7 at 09:40, v6 at 09:45. There is no shared "this stock just blew up — sit out" memo across variants. Each runs in its own process and reads the scorer fresh.

> **WHY THIS MULTIPLIES LOSSES**
>
> This is also exactly why "running 8 variants" today produced "1 variant's loss × 7" instead of 8 different bets. They are not diversified strategies — they are 7 copies of the same prior with slight risk-management differences.

4. Q3 — Why SIDEWAYS Produced 28 LONG-Only Trades

HEADLINE FINDING

The regime-aware slot partition (`15 LONG / 5 SHORT` for SIDEWAYS) **was disabled in v5_8** as part of an experiment, and **never enforced in v6** because v6 bypasses the slot logic entirely. Engines that kept the partition (v5_6, v5_7) still went 18–23 LONG vs 1 SHORT — the regime classifier marked the day SIDEWAYS but the scorer's underlying long bias dominated.

The regime partition (defined but inconsistently applied)

From `prototype/v5/risk_manager.py:55-59` :

Regime	LONG slots	SHORT slots
BULL	18	2
SIDEWAYS	**15**	**5**
BEAR	8	12

What actually happened today

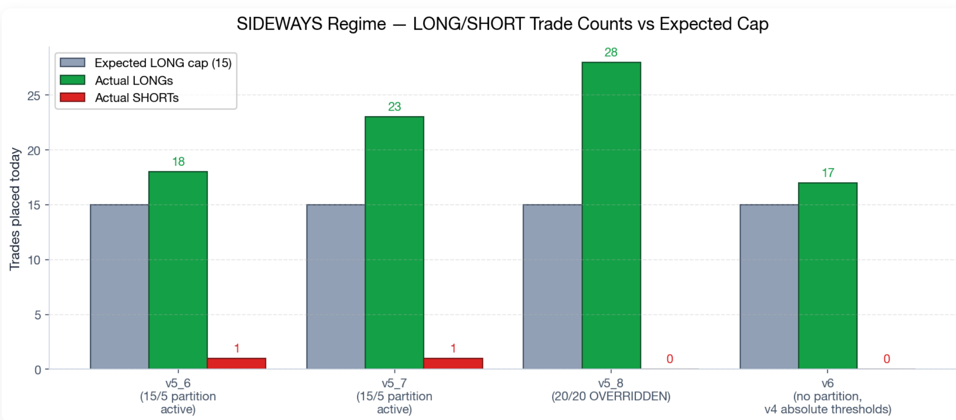

Figure 6 · Expected SIDEWAYS LONG cap (15) vs actual LONGs/SHORTs. v5_6/v5_7 honored the partition but the scorer never produced 5 SHORT candidates. v5_8 explicitly disabled it. v6 doesn't use it.

Per-variant root cause

Variant	Trades	LONG/SHORT	Why
`v5_6`	19	18 / 1	Partition active, but scorer surfaced no SHORTs in green sectors → drifted long-heavy
`v5_7`	24	23 / 1	Same as v5_6 — partition is a cap, not a quota
`v5_8`	28	28 / 0	**Partition explicitly overridden to 20/20** in test script (line 47–50)
`v6`	17	17 / 0	Bypasses partition — uses absolute v4 thresholds (BUY≥60, SHORT≤35)

THE DEEPER FINDING

Even when the partition is active (v5_6, v5_7), it is a **ceiling**, not a **floor**. If the scorer doesn't produce 5 SHORT candidates with the right scores, you don't get 5 SHORTs — you get 0–1. SIDEWAYS regime needs a **SHORT-floor** rule, not just a SHORT-ceiling, to actually neutralise direction bias.

5. Q4 — Where v5_8 and v6 Actually Live

Memory said you have prototype dirs `v5`, `v5_2`, `v5_3`, `v5_6`, `v5_7`. There are no `v5_8` or `v6` dirs. Yet both produce daily reports. Mystery solved:

Variant	Lives at	Reuses	Mechanism
`v5_8`	`scripts/v5_8-paper-trade.py`	`prototype/v5/*`	Imports v5 risk manager, monkey-patches `REGIME_SLOT_SPLIT` in-memory before calling `generate_signals()`
`v6`	`scripts/v6-paper-trade.py`	`prototype/v4/composite_scorer`	Skips v5 entirely; calls v4 scorer with hard-coded absolute thresholds (`V6_BUY_MIN_SCORE=60`, `V6_SHORT_MAX_SCORE=35`)

WHAT THIS MEANS FOR THE LAB

You have **2 real engines** (v4 scorer + v5 signal_engine) and **5 dispatcher scripts** wrapping them. The "8 variants" mental model is misleading — they share roots. Pruning is easier than it looks: delete v5_8/v6 dispatcher scripts and you've removed 2 of 8 "engines" without losing any unique strategy.

6. Recommendations & Immediate Actions

Tonight (before tomorrow's open)

#	Action	Why	Effort
1	Add VEDL to a temporary `blacklist.json` through 2026-05-07	Engines won't re-enter on the demerger noise that will continue this week	15 min
2	Manually adjust today's EOD report — strip VEDL from each variant's P&L	Tomorrow's running totals would otherwise carry a fictional loss	10 min
3	Add this incident to `RETIRED_ENGINES.md` learning notes	Future Soumya will hit this again — corp actions repeat quarterly	5 min

This weekend (Phase 1 hygiene fix)

#	Action	Where	Effort
4	Add corporate-action ex-date filter to `data_engine.py`	Pull NSE corp-actions JSON daily; skip stocks with ex-date today	~3 hr
5	Promote v5_2's per-trade kill-switch (Rs 5K) to `risk_manager.py` baseline	So all 6 variants inherit it, not just v5_2	~1 hr
6	Add per-position daily SL — e.g. cap any single ticker loss at 0.5% of portfolio (Rs 25K)	The missing rail in Figure 3	~2 hr

Next sprint (architecture)

#	Action	Why	Effort
7	Add per-stock cool-off — if any variant stoplosses ticker X, others sit out X for 2 hours	Stops the convergent-loss multiplier (Q2)	~4 hr
8	Convert SHORT-slot partition from ceiling to **floor** in SIDEWAYS regime	Forces direction-neutrality; today's 28 LONG / 0 SHORT outcome wouldn't repeat	~3 hr
9	Prune the lab: retire v5_8 + v6 dispatcher scripts after their experiments conclude	"8 engines" is misleading — 2 real, 6 wrappers	discussion

Where to source the corp-action calendar

Source	What it gives	Cost
NSE official feed	`nseindia.com/api/corporates-corporateActions` — daily JSON	Free
BSE corporate actions	Daily JSON via bseindia.com	Free
Zerodha Kite Connect	Has corp-actions endpoint after Phase 3	Rs 2K/mo
nsetools / nsepython	Python wrappers around NSE API	Free (rate-limited)

THE BIGGER PATTERN

This is the 3rd hidden failure mode the side-by-side observation window has exposed in 3 days: 04-28 SHORTs forced in green tapes, 04-29 slot partition starves LONGs, 04-30 zero corp-action filter. Running 7 engines on the same data is exactly what surfaced it — strategy losses look different across variants; **data losses bleed identically**. That is the most valuable signal of this entire observation period.

Closing the Loop

What this document is

A root-cause analysis of the 2026-04-30 paper-trading loss across all 7 active TradePilot variants. Generated from engine code (`engine/src/risk/mod.rs`, `prototype/v5/risk_manager.py`, dispatcher scripts) and EOD reports under `docs/paper-trades/`.

What changes next

Three Phase-1 hygiene fixes (corp-action filter, baseline kill-switch, per-position SL) and three architecture improvements (cool-off, SHORT floor, lab pruning). All actions are listed in §6 with effort estimates.

What this is not

This is not a verdict on any individual engine's strategy. The −Rs 25K adjusted loss across all 7 engines on a sideways tape is in normal range. The story today is about the **data layer**, not the **strategy layer**.